Name

Mathematics Extension 1 Year 11

Further Trigonometric Equations

Book 3

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Syllabus Content

ME1-11-03 solves problems in three dimensions using trigonometry and simplifies expressions, proves results and solves problems involving compound angles using trigonometric identities

Further trigonometric equations

- Solve trigonometric equations involving factorisation and/or substitution of trigonometric identities over restricted domains
- Examine the representations of $a\ \cos\ x + b\sin x$ as $R\cos(x \pm \alpha)$ or $R\sin(x \pm \alpha)$ with and without graphing applications, where $a \neq 0$ and $b \neq 0$
- Apply the representations of $a\ \cos\ x + b\sin x$ as $R\cos(x \pm \alpha)$ or $R\sin(x \pm \alpha)$ to graph functions, solve equations of the form $a\cos x + b\sin x = c$ over restricted domains, and model and solve related problems
- Interpret solutions of trigonometric equations graphically with and without graphing applications
- Model and solve practical problems using trigonometric equations and analyse their solutions in a variety of contexts

Review of Trigonometric Identities and Equations

On Reference Sheet

Reciprocal Trigonometric Ratios

$$\sec A = \frac{1}{\cos A}, \cos A \neq 0$$

$$\operatorname{cosec} A = \frac{1}{\sin A}, \sin A \neq 0$$

$$\cot A = \frac{\cos A}{\sin A}, \sin A \neq 0$$

Pythagorean Identity

$$\cos^2 x + \sin^2 x = 1$$

Compound Angle Identities

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

Not on Reference Sheet

Tangent and Cotangent Ratio

$$\tan A = \frac{\sin A}{\cos A} \qquad \cot A = \frac{1}{\tan A}$$

Complementary Angle Identities

$$\sin(90° - \theta) = \cos\theta \qquad \cos(90° - \theta) = \sin\theta$$

$$\tan(90° - \theta) = \cot\theta \qquad \cot(90° - \theta) = \tan\theta$$

$$\sec(90° - \theta) = \operatorname{cosec}\theta \qquad \operatorname{cosec}(90° - \theta) = \sec\theta$$

Symmetry

$$\sin(-x) = -\sin x \qquad \tan(-x) = -\tan x \qquad \cos(-x) = \cos x$$

Pythagorean Identity

$$1 + \tan^2 x = \sec^2 x$$

$$\cot^2 x + 1 = \operatorname{cosec}^2 x$$

Compound Angle Identities

$$\sin(A - B) = \sin A \cos B - \cos A \sin B$$

$$\cos(A - B) = \cos A \cos B + \sin A \sin B$$

$$\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

Double Angle Identities

$$\sin(2A) = 2 \sin A \cos A$$

$$\cos(2A) = \cos^2 A - \sin^2 A$$

$$= 2\cos^2\theta - 1$$

$$= 1 - 2\sin^2\theta$$

$$\tan(2A) = \frac{2\tan A}{1 - \tan^2 A}$$

1 Expand these compound angle expressions.

a $\sin\left(x+\frac{\pi}{4}\right)+\sin\left(x-\frac{\pi}{4}\right)$

$\sqrt{2}\sin x$

b $\cos\left(x+\frac{\pi}{3}\right)+\cos\left(x-\frac{\pi}{3}\right)$

$\cos x$

c $\sin\left(x+\frac{\pi}{6}\right)-\sin\left(x-\frac{\pi}{6}\right)$

$\cos x$

d $\sin\left(x+\frac{\pi}{6}\right)$

$\left(\frac{\sqrt{3}}{2}\right)\sin x+(½)\cos x$

e $\cos\left(x-\frac{\pi}{4}\right)$

$\left(\frac{1}{\sqrt{2}}\right)\cos x+\left(\frac{1}{\sqrt{2}}\right)\sin x$

f $2\sin\left(x+\frac{\pi}{3}\right)$

$\sqrt{3}\sin x+\cos x$

2 Use double-angle rule to simplify these expressions such there is no squared term.

a $2\sin x\cos x$

$\sin 2x$

b $\cos^2 x-\sin^2 x$

$\cos 2x$

c $2\cos^2 x-1$

$\cos 2x$

d $1-2\sin^2 x$

$\cos 2x$

e $2\sin 2x\cos 2x$

$\sin 4x$

f $2\sin 3x\cos 3x$

$\sin 6x$

g $\cos^2 2x-\sin^2 2x$

$\cos 4x$

h $2\cos^2 3x-1$

$\cos 6x$

i $1-2\sin^2\left(\frac{x}{2}\right)$

$\cos x$

j $2\sin\left(\frac{x}{2}\right)\cos\left(\frac{x}{2}\right)$

$\sin x$

k $\sin x\cos x$

$½\sin 2x$

l $4\sin x\cos x$

$2\sin 2x$

m $6\sin 2x\cos 2x$

$3\sin 4x$

n $\sin^2 x-\cos^2 x$

$-\cos 2x$

o $\frac{1}{2}(1-2\sin^2 x)$

$½\cos 2x$

p $\cos^2 x-\frac{1}{2}$

$½\cos 2x$

q $\frac{1}{2}-\sin^2 x$

$½\cos 2x$

r $4\cos^2 x-2$

$2\cos 2x$

3 Use the Pythagorean identity to rewrite these expressions in terms of $\cos^2 x$.

a $1 - \sin^2 x + \cos^2 x$

$2\cos^2 x$

b $5\cos^2 x - 5\sin^2 x + 5$

$10\cos^2 x$

c $1 - \sin^2 x + 4\cos^2 x$

$5\cos^2 x$

4 Use the Pythagorean identity to rewrite these expressions in terms of $\sin^2 x$.

a $2\sin^2 x - 2\cos^2 x + 2$

$4\sin^2 x$

b $5 - 5\cos^2 x + \sin^2 x$

$6\sin^2 x$

c $2\sin^2 x + 1 - \cos^2 x$

$3\sin^2 x$

5 Solve these trigonometric equations for $x \in [0, 2\pi)$

a $\sin x = \frac{1}{2}$

$\frac{\pi}{6}, \frac{5\pi}{6}$

b $\sin x = \frac{1}{\sqrt{2}}$

$\frac{\pi}{4}, \frac{3\pi}{4}$

c $\cos x = \frac{\sqrt{3}}{2}$

$\frac{\pi}{6}, \frac{11\pi}{6}$

d $\cos x = \frac{1}{\sqrt{2}}$

$\frac{\pi}{4}, \frac{7\pi}{4}$

e $\tan x = \frac{1}{\sqrt{3}}$

$\frac{\pi}{6}, \frac{7\pi}{6}$

f $\sin x = -\frac{1}{2}$

$\frac{7\pi}{6}, \frac{11\pi}{6}$

g $\sin x = -\frac{1}{\sqrt{2}}$

$\frac{5\pi}{4}, \frac{7\pi}{4}$

h $\cos x = -\frac{\sqrt{3}}{2}$

$\frac{5\pi}{6}, \frac{7\pi}{6}$

i $\tan x = -1$

$\frac{3\pi}{4}, \frac{7\pi}{4}$

j $\cos x = -1$

π

k $2 \sin x = 1$

$\frac{\pi}{6}, \frac{5\pi}{6}$

l $\sqrt{2}\sin x = 1$

$\frac{\pi}{4}, \frac{3\pi}{4}$

m $2 \cos x + \sqrt{3} = 0$

$\frac{5\pi}{6}, \frac{7\pi}{6}$

n $3 \sin x = \frac{3}{2}$

$\frac{\pi}{6}, \frac{5\pi}{6}$

o $2 \sin x - \sqrt{2} = 0$

$\frac{\pi}{4}, \frac{3\pi}{4}$

6 Factorise

a $\sin x + \sin x \cos x$

$\sin x(1 + \cos x)$

b $\cos x - \sin x \cos x$

$\cos x(1 - \sin x)$

c $\sin^2 x - \sin x$

$\sin x(\sin x - 1)$

d $\cos^2 x + \cos x$

$\cos x(\cos x + 1)$

e $2\sin^2 x + \sin x - 1$

$(2\sin x - 1)(\sin x + 1)$

f $\sin^2 x - \cos^2 x$

$(\sin x - \cos x)(\sin x + \cos x) = -\cos 2x$

7 Solve these trigonometric equations for $x \in [0,2\pi)$

a $\sin 2x = \frac{1}{2}$

$\frac{\pi}{12}, \frac{5\pi}{12}, \frac{13\pi}{12}, \frac{17\pi}{12}$

b $\cos 2x = \frac{1}{2}$

$\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$

c $\tan 2x = 1$

$\frac{\pi}{8}, \frac{5\pi}{8}, \frac{9\pi}{8}, \frac{13\pi}{8}$

d $\cos 2x = 0$

$\frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$

e $\tan 2x = \sqrt{3}$

$\frac{\pi}{6}, \frac{2\pi}{3}, \frac{7\pi}{6}, \frac{5\pi}{3}$

f $\cos\left(\frac{x}{2}\right) = \frac{\sqrt{3}}{2}$

$\pi/3$

8 Solve these trigonometric equations for $x \in [0,2\pi)$

a $\sin\left(x - \frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$

$\frac{\pi}{2}, \pi$

b $\cos\left(x - \frac{\pi}{6}\right) = -\frac{1}{2}$

$\frac{5\pi}{6}, \frac{3\pi}{2}$

c $\sin\left(x + \frac{\pi}{4}\right) = 0$

$\frac{3\pi}{4}, \frac{7\pi}{4}$

d $\sin\left(x - \frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$

$\frac{2\pi}{3}, \pi$

e $\tan\left(x - \frac{\pi}{6}\right) = -1$

$\frac{11\pi}{12}, \frac{23\pi}{12}$

9 Prove:

a $\sin 2x + 2\sin x = 2\sin x\ (\cos x + 1)$

$\text{LHS} = \sin 2x + 2\sin^2 x$
$= 2\sin x \cos x + 2\sin x$
$= 2\sin x\ (\cos x + 1)$
$= \text{RHS}$

b $\cos 2x + \cos^2 x = 3\cos^2 x - 1$

$\text{LHS} = \cos 2x + \cos^2 x$
$= (2\cos^2 x - 1) + \cos^2 x \text{ (using } \cos 2x = 2\cos^2 x - 1)$
$= 3\cos^2 x - 1$
$= \text{RHS}$

c $\cos 2x - \sin^2 x = 1 - 3\sin^2 x$

$\text{LHS} = \cos 2x - \sin^2 x$
$= (1 - 2\sin^2 x) - \sin^2 x \text{ (using } \cos 2x = 1 - 2\sin^2 x)$
$= 1 - 3\sin^2 x$
$= \text{RHS}$

d $\dfrac{\sin 2x}{\sin x} = 2\cos x$

$\text{LHS} = \dfrac{\sin 2x}{\sin x}$
$= \dfrac{2\sin x \cos x}{\sin x}$
$= 2\cos x$
$= \text{RHS}$

e $\dfrac{\sin 2x}{(1 + \cos 2x)} = \tan x$

$\text{LHS} = (\sin 2x)/(1 + \cos 2x)$
$= (2\sin x \cos x)/(1 + 2\cos^2 x - 1)$
$= (2\sin x \cos x)/(2\cos^2 x)$
$= (\sin x)/(\cos x)$
$= \tan x$
$= \text{RHS}$

f $\dfrac{(1 - \cos 2x)}{\sin x} = 2\sin x$

$\text{LHS} = \dfrac{1 - \cos 2x}{\sin x}$
$= \dfrac{1 - (1 - 2\sin^2 x)}{\sin x}$
$= \dfrac{2\sin^2 x}{\sin x}$
$= 2\sin x$
$= \text{RHS}$

10 Rewrite the following such that there is only a single angle and trigonometric function.

a $\cos 2x + 3\cos^2 x$

$5\cos^2 x - 1$

b $\sin 2x - 4\sin x \cos x$

$-\sin 2x$

c $\sin^2 x + \sin 2x + \cos^2 x$

$1 + \sin 2x$

d $\dfrac{1 - \cos 2x}{1 + \cos 2x}$

$\tan^2 x$

e $\dfrac{\sin 2x}{1 - \cos 2x}$

$\cot x$

f $(1 + \tan^2 x)\cos^2 x$

1

11 Solve these trigonometric equations for $x \in [0,2\pi)$

a $\cos^2 x = \frac{1}{2}$

$\frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$

b $\tan^2 x = 1$

$\frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$

c $2\cos^2 x = 1$

$\frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$

d $4\cos^2 x - 3 = 0$

$\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$

e $\sin^2 x - \sin x = 0$

$0, \frac{\pi}{2}, \pi$

f $\cos^2 x - \cos x = 0$

$0, \frac{\pi}{2}, \frac{3\pi}{2}$

12 Solve these trigonometric equations for $x \in [0,2\pi)$

a $\sin x = \cos x$

$\frac{\pi}{4}, \frac{5\pi}{4}$

b $\sqrt{3}\sin x = \cos x$

$\frac{\pi}{6}, \frac{7\pi}{6}$

c $\sin x + \cos x = 0$

$\frac{3\pi}{4}, \frac{7\pi}{4}$

d $\sin 2x = \cos 2x$

$\frac{\pi}{8}, \frac{5\pi}{8}, \frac{9\pi}{8}, \frac{13\pi}{8}$

13 Solve these trigonometric equations for $x \in [0,2\pi)$

a $2\sin^2 x - \sin x - 1 = 0$

$\frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$

b $2\sin^2 x + 3\sin x + 1 = 0$

$\frac{7\pi}{6}, \frac{3\pi}{2}, \frac{11\pi}{6}$

c $\sin^2 x + \sin x - 2 = 0$

$\frac{\pi}{2}$

d $\cos^2 x - 2\cos x - 3 = 0$

π

Solving Trigonometric Equations by using Identities

General Strategy for Solving Trigonometric Equations

When a trigonometric equation has **different trigonometric functions** or **different angles**, use the trigonometric identities to:

1. Make all the angles the same.
2. Make all the functions the same.

Example Solve trigonometric equations using identities

Solve $\cos 2x = \sin x$ for $x \in [0,2\pi)$

$$1 - 2\sin^2 x = \sin x$$

$$2\sin^2 x + \sin x - 1 = 0$$

$$(2\sin x - 1)(\sin x + 1) = 0$$

$$\sin x = \frac{1}{2} \text{ or } -1$$

$$\therefore x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$$

We used the identity $\cos 2x =$ ..... to solve this equation.

This form of the identity was useful because it transformed the equation into a q..... equation involving only $\sin x$.

Guided Practice

Solve for $x \in [0, 2\pi)$

a $\cos 2x + \cos x = 0$

$\frac{\pi}{3}, \pi, \frac{5\pi}{3}$

b $2\cos^2 x + \sin x - 1 = 0$

$\frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$

c $\sin 2x = \sin x$

$0, \frac{\pi}{3}, \pi, \frac{5\pi}{3}$

d $2\tan\theta = \sec\theta$

$\theta = \frac{\pi}{6}, \frac{5\pi}{6}$

Identify homogeneous trigonometric equations

Homogeneous trigonometric equations contain terms where the powers of sin x and cos x add to the same total.

✗	✓	✓	✗
$\sin^2 x + \sin x = 0$	$\sin^2 x + \sin x \cos x = 0$	$3\sin^4 x - 4\cos^4 x = 2$	$\sin^4 x + \sin^2 x \cos x = 2$

Strategies for Solving Homogeneous Equations

To solve, divide every term by a power of $\cos x$ to get a polynomial in $\tan x$.

⚠ When you divide by $\cos x$, you are assuming that $\cos x \neq 0$.
Therefore, you must **always** check whether $x = \frac{\pi}{2}$ or $\frac{3\pi}{2}$ is a solution by substituting.

⚠ Remember that $\sin 2x$ and $\cos 2x$ can be expanded to give terms of degree 2.

Example Solve homogeneous trigonometric equations

Solve $\sin^2 x - \sin x \cos x = 0$ for $x \in [0, 2\pi)$

$$\div \cos^2 x \qquad \div \cos^2 x$$

$$\tan^2 x - \tan x = 0$$

$$\tan x (\tan x - 1) = 0$$

$$\tan x = 0 \text{ or } 1$$

$$\therefore x = 0, \pi, \frac{\pi}{4}, \frac{5\pi}{4}$$

Check whether $x = \frac{\pi}{2}, \frac{3\pi}{2}$ is a solution by substitution

$$\sin^2\left(\frac{\pi}{2}\right) - \sin\left(\frac{\pi}{2}\right)\cos\left(\frac{\pi}{2}\right) = 1 \neq 0$$

$\therefore x = 0, \pi, \frac{\pi}{4}, \frac{5\pi}{4}$ are the only solutions.

Guided Practice

Solve for $x \in [0,2\pi)$

a $\sqrt{3}\sin x = \cos x$

$\frac{\pi}{6}, \frac{7\pi}{6}$

b $\sin x \cos x - \sqrt{3}\cos^2 x = 0$

$\frac{\pi}{3}, \frac{\pi}{2}, \frac{4\pi}{3}, \frac{3\pi}{2}$

c $\sin x \cos x + \cos^2 x = 0$

$\frac{\pi}{2}, \frac{3\pi}{4}, \frac{3\pi}{2}, \frac{7\pi}{4}$

d $\sin^2 x + \sin 2x - \cos^2 x = 0$

$\frac{\pi}{8}, \frac{5\pi}{8}, \frac{9\pi}{8}, \frac{13\pi}{8}$

Squaring Both Sides

Squaring both sides lets you apply a double-angle identity to simplify the equation.
There is *usually* a better method you could use.

⚠ Squaring introduces extra solutions, so always check your answers by substituting them into the original equation.

Example Solve trigonometric equations by squaring both sides

Solve $\sin x + \cos x = 1$ for $x \in [0,2\pi)$

$$(\sin x + \cos x)^2 = 1^2$$
$$\sin^2 x + 2\sin x \cos x + \cos^2 x = 1$$
$$2\sin x \cos x = 0$$
$$\sin 2x = 0$$
$$2x = 0, \pi, 2\pi, 3\pi$$
$$x = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$$

Checking solutions:

$\sin 0 + \cos 0 = 1$ ✓ $\sin\frac{\pi}{2} + \cos\frac{\pi}{2} = 1$ ✓ $\sin\pi + \cos\pi = -1$ ✗ $\sin\frac{3\pi}{2} + \cos\frac{3\pi}{2} = -1$ ✗

$\therefore x = 0, \frac{\pi}{2}$ are the only solutions

Guided Practice

Solve by squaring both sides for $x \in [0,2\pi)$

a $\sec x - \tan x = \sqrt{3}$

$\frac{11\pi}{6}$

b $\sin x + \cos x = \sqrt{2}$

$x = \frac{\pi}{4}$

1 Consider the equation $\sin 2x - \cos x = 0$.

a By using a double-angle formula and then factoring, show that $\cos x = 0$ or $\sin x = \frac{1}{2}$.

b Hence solve the equation for $0 \leq x \leq 2\pi$.

$$\begin{aligned}
\sin 2x - \cos x &= 0 \\
2\sin x \cos x - \cos x &= 0 \\
\cos x(2\sin x - 1) &= 0 \\
\therefore \cos x = 0 &\text{ or } \sin x = \frac{1}{2} \\
\cos x = 0 &: x = \frac{\pi}{2}, \frac{3\pi}{2} \\
\sin x = \frac{1}{2} &: x = \frac{\pi}{6}, \frac{5\pi}{6} \\
\therefore x &= \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6} \text{ or } \frac{3\pi}{2}
\end{aligned}$$

2 Consider the equation $\cos 2x - \cos x = 0$.

a By using a double-angle formula and then factoring, show that $\cos x = 1$ or $-\frac{1}{2}$.

b Hence solve the equation for $0 \leq x \leq 2\pi$.

$$\begin{aligned}
\cos 2x - \cos x &= 0 \\
2\cos^2 x - 1 - \cos x &= 0 \\
2\cos^2 x - \cos x - 1 &= 0 \\
(2\cos x + 1)(\cos x - 1) &= 0 \\
\therefore \cos x = 1 &\text{ or } -\frac{1}{2} \\
\cos x = 1 &: x = 0, 2\pi \\
\cos x = -\frac{1}{2} &: x = \frac{2\pi}{3}, \frac{4\pi}{3} \\
\therefore x &= 0, \frac{2\pi}{3}, \frac{4\pi}{3} \text{ or } 2\pi
\end{aligned}$$

3 Consider the equation $\sin\left(x + \frac{\pi}{4}\right) = 2\cos\left(x - \frac{\pi}{4}\right)$.

a Use compound-angle formulae to show that $\tan x = -1$.

b Hence solve the equation for $0 \leq x \leq 2\pi$.

$$\begin{aligned}
\sin x \cos\left(\frac{\pi}{4}\right) + \cos x \sin\left(\frac{\pi}{4}\right) &= 2\left(\cos x \cos\left(\frac{\pi}{4}\right) + \sin x \sin\left(\frac{\pi}{4}\right)\right) \\
\left(\frac{1}{\sqrt{2}}\right)\sin x + \left(\frac{1}{\sqrt{2}}\right)\cos x &= \left(\frac{2}{\sqrt{2}}\right)\cos x + \left(\frac{2}{\sqrt{2}}\right)\sin x \\
\sin x + \cos x &= 2\cos x + 2\sin x \\
-\sin x &= \cos x \\
\tan x &= -1 \\
\therefore x &= \frac{3\pi}{4} \text{ or } \frac{7\pi}{4}
\end{aligned}$$

4 Solve $2\tan\theta = \sec\theta$, for $0 \leq \theta \leq 360°$ by:

a Applying identities

$$\begin{aligned}
\frac{2\sin\theta}{\cos\theta} &= \frac{1}{\cos\theta} \\
\sin\theta &= \frac{1}{2} \\
\theta &= 30° \text{ or } 150°
\end{aligned}$$

b Squaring both sides then using identities

$$\begin{aligned}
4\tan^2\theta &= \sec^2\theta \\
4\sec^2\theta - 4 &= \sec^2\theta \\
\sec^2\theta &= \frac{4}{3} \\
\cos\theta &= \pm\frac{\sqrt{3}}{2} \\
\theta &= 30°, 150°, 210°, 330°
\end{aligned}$$

After verifying solutions, only solutions are $\theta = 30°, 150°$

5 Use compound-angle formulae to solve, for $0 \leq \theta \leq 2\pi$:

a $\sin\left(\theta + \frac{\pi}{6}\right) = 2\sin\left(\theta - \frac{\pi}{6}\right)$

$$\sin\theta\cos\left(\frac{\pi}{6}\right) + \cos\theta\sin\left(\frac{\pi}{6}\right) = 2\left(\sin\theta\cos\left(\frac{\pi}{6}\right) - \cos\theta\sin\left(\frac{\pi}{6}\right)\right)$$
$$\left(\frac{\sqrt{3}}{2}\right)\sin\theta + \left(\frac{1}{2}\right)\cos\theta = \sqrt{3}\sin\theta - \cos\theta$$
$$\left(\frac{3}{2}\right)\cos\theta = \left(\frac{\sqrt{3}}{2}\right)\sin\theta$$
$$\tan\theta = \sqrt{3}$$
$$\therefore \theta = \frac{\pi}{3} \text{ or } \frac{4\pi}{3}$$

b $\cos\left(\theta - \frac{\pi}{6}\right) = 2\cos\left(\theta + \frac{\pi}{6}\right)$

$$\cos\theta\cos\left(\frac{\pi}{6}\right) + \sin\theta\sin\left(\frac{\pi}{6}\right) = 2\left(\cos\theta\cos\left(\frac{\pi}{6}\right) - \sin\theta\sin\left(\frac{\pi}{6}\right)\right)$$
$$\left(\frac{\sqrt{3}}{2}\right)\cos\theta + \left(\frac{1}{2}\right)\sin\theta = \sqrt{3}\cos\theta - \sin\theta$$
$$\left(\frac{3}{2}\right)\sin\theta = \left(\frac{\sqrt{3}}{2}\right)\cos\theta$$
$$\tan\theta = \frac{1}{\sqrt{3}}$$
$$\therefore \theta = \frac{\pi}{6} \text{ or } \frac{7\pi}{6}$$

c $\cos 4\theta\cos\theta + \sin 4\theta\sin\theta = \frac{1}{2}$

$$\cos(4\theta - \theta) = \frac{1}{2}, \text{using } \cos(A - B) = \cos A\cos B + \sin A\sin B$$
$$\cos 3\theta = \frac{1}{2}$$
$$3\theta = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3}, \frac{13\pi}{3}, \frac{17\pi}{3}$$
$$\therefore \theta = \frac{\pi}{9}, \frac{5\pi}{9}, \frac{7\pi}{9}, \frac{11\pi}{9}, \frac{13\pi}{9} \text{ or } \frac{17\pi}{9}$$

d $\cos 3\theta = \cos 2\theta\cos\theta$
(Hint: write $\cos 3\theta$ as $\cos(2\theta + \theta)$)

$$\cos(2\theta + \theta) = \cos 2\theta\cos\theta$$
$$\cos 2\theta\cos\theta - \sin 2\theta\sin\theta = \cos 2\theta\cos\theta$$
$$\sin 2\theta\sin\theta = 0$$
$$2\sin^2\theta\cos\theta = 0$$
$$\sin\theta = 0 \text{ or } \cos\theta = 0$$
$$\therefore \theta = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2} \text{ or } 2\pi$$

6 Use double-angle formulae to solve, for $0 \leq x \leq 2\pi$:

a $\sin 2x = \sin x$

$$2\sin x\cos x = \sin x$$
$$\sin x(2\cos x - 1) = 0$$
$$\sin x = 0 \text{ or } \cos x = \frac{1}{2}$$
$$\therefore x = 0, \frac{\pi}{3}, \pi, \frac{5\pi}{3} \text{ or } 2\pi$$

b $\sin 2x + \sqrt{3}\cos x = 0$

$$2\sin x\cos x + \sqrt{3}\cos x = 0$$
$$\cos x(2\sin x + \sqrt{3}) = 0$$
$$\cos x = 0 \text{ or } \sin x = -\frac{\sqrt{3}}{2}$$
$$\therefore x = \frac{\pi}{2}, \frac{4\pi}{3}, \frac{3\pi}{2} \text{ or } \frac{5\pi}{3}$$

c $3\sin x + \cos 2x = 2$

$$3\sin x + 1 - 2\sin^2 x = 2$$
$$2\sin^2 x - 3\sin x + 1 = 0$$
$$(2\sin x - 1)(\sin x - 1) = 0$$
$$\sin x = \frac{1}{2} \text{ or } 1$$
$$\therefore x = \frac{\pi}{6}, \frac{\pi}{2} \text{ or } \frac{5\pi}{6}$$

d $\cos 2x + 3\cos x + 2 = 0$

$$2\cos^2 x - 1 + 3\cos x + 2 = 0$$
$$2\cos^2 x + 3\cos x + 1 = 0$$
$$(2\cos x + 1)(\cos x + 1) = 0$$
$$\cos x = -\frac{1}{2} \text{ or } -1$$
$$\therefore x = \frac{2\pi}{3}, \pi \text{ or } \frac{4\pi}{3}$$

e $\tan 2x + \tan x = 0$

$$\frac{2\tan x}{1 - \tan^2 x} + \tan x = 0$$
$$2\tan x + \tan x(1 - \tan^2 x) = 0$$
$$\tan x(2 + 1 - \tan^2 x) = 0$$
$$\tan x(3 - \tan^2 x) = 0$$
$$\tan x = 0 \text{ or } \tan x = \pm\sqrt{3}$$
$$\therefore x = 0, \frac{\pi}{3}, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}, \frac{5\pi}{3} \text{ or } 2\pi$$

f $\sin 2x = \tan x$

$$2\sin x\cos x = \frac{\sin x}{\cos x}$$
$$2\sin x\cos^2 x = \sin x$$
$$\sin x(2\cos^2 x - 1) = 0$$
$$\sin x = 0 \text{ or } \cos^2 x = \frac{1}{2}$$
$$\sin x = 0 : x = 0, \pi, 2\pi$$
$$\cos x = \pm\frac{1}{\sqrt{2}} : x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$$

7 Solve, for $0° \le \theta \le 360°$, giving solutions correct to the nearest minute where necessary:

a $2\cos 2\theta + 4\cos\theta = 1$

$$
\begin{aligned}
2(2\cos^2\theta - 1) + 4\cos\theta &= 1\\
4\cos^2\theta + 4\cos\theta - 3 &= 0\\
(2\cos\theta + 3)(2\cos\theta - 1) &= 0\\
\cos\theta &= \frac{1}{2}\left(\cos\theta = -\frac{3}{2}\text{rejected}\right)\\
\therefore\ \theta &= 60° \text{ or } 300°
\end{aligned}
$$

b $2\cos^2\theta + \cos 2\theta = 0$

$$
\begin{aligned}
2\cos^2\theta + 2\cos^2\theta - 1 &= 0\\
4\cos^2\theta &= 1\\
\cos\theta &= \pm\frac{1}{2}\\
\therefore\ \theta &= 60°, 120°, 240° \text{ or } 30
\end{aligned}
$$

c $3\cos 2\theta + \sin\theta = 1$

$$
\begin{aligned}
3(1 - 2\sin^2\theta) + \sin\theta &= 1\\
6\sin^2\theta - \sin\theta - 2 &= 0\\
(3\sin\theta - 2)(2\sin\theta + 1) &= 0\\
\sin\theta &= \frac{2}{3} \text{ or } -\frac{1}{2}\\
\sin\theta &= \frac{2}{3}: \theta = 41°49', 138°11'\\
\sin\theta &= -\frac{1}{2}: \theta = 210°, 330°\\
\therefore\ \theta &= 41°49', 138°11', 210° \text{ or } 330°
\end{aligned}
$$

d $\cos 2\theta = 3\cos^2\theta - 2\sin^2\theta$

$$
\begin{aligned}
\cos^2\theta - \sin^2\theta &= 3\cos^2\theta - 2\sin^2\theta\\
2\cos^2\theta &= \sin^2\theta\\
\tan^2\theta &= 2\\
\tan\theta &= \pm\sqrt{2}\\
\therefore\ \theta &= 54°44', 125°16', 234°44' \text{ or } 305°16'
\end{aligned}
$$

e $10\cos\theta + 13\cos\left(\frac{\theta}{2}\right) = 5$

$$
\begin{aligned}
10\cos\theta + 13\cos\left(\frac{\theta}{2}\right) &= 5\\
10\left(2\cos^2\left(\frac{\theta}{2}\right) - 1\right) + 13\cos\left(\frac{\theta}{2}\right) &= 5\\
20\cos^2\left(\frac{\theta}{2}\right) + 13\cos\left(\frac{\theta}{2}\right) - 15 &= 0\\
\left(4\cos\left(\frac{\theta}{2}\right) + 5\right)\left(5\cos\left(\frac{\theta}{2}\right) - 3\right) &= 0\\
\cos\left(\frac{\theta}{2}\right) &= \frac{3}{5}\left(\cos\left(\frac{\theta}{2}\right) = -\frac{5}{4}\text{rejected}\right)\\
\frac{\theta}{2} &= 53°8'\\
\therefore\ \theta &= 106°16'
\end{aligned}
$$

f $\tan\theta = 3\tan\left(\frac{\theta}{2}\right)$

$$
\begin{aligned}
\frac{2\tan\left(\frac{\theta}{2}\right)}{1 - \tan^2\left(\frac{\theta}{2}\right)} &= 3\tan\left(\frac{\theta}{2}\right)\\
2\tan\left(\frac{\theta}{2}\right) &= 3\tan\left(\frac{\theta}{2}\right)\left(1 - \tan^2\left(\frac{\theta}{2}\right)\right)\\
\tan\left(\frac{\theta}{2}\right)\left(2 - 3 + 3\tan^2\left(\frac{\theta}{2}\right)\right) &= 0\\
\tan\left(\frac{\theta}{2}\right)\left(3\tan^2\left(\frac{\theta}{2}\right) - 1\right) &= 0\\
\tan(\theta/2) &= 0 \text{ or } \tan(\theta/2) = \pm\frac{1}{\sqrt{3}}\\
\tan\left(\frac{\theta}{2}\right) &= 0: \theta = 0°, 360°\\
\tan\left(\frac{\theta}{2}\right) &= \frac{1}{\sqrt{3}}: \frac{\theta}{2} = 30°, \theta = 60°\\
\tan\left(\frac{\theta}{2}\right) &= -\frac{1}{\sqrt{3}}: \frac{\theta}{2} = 150°, \theta = 300°\\
\therefore\ \theta &= 0°, 60°, 300° \text{ or } 360°
\end{aligned}
$$

g $\cos^2 2\theta = \sin^2\theta$

(Hint: Use $\sin^2\theta = \frac{1}{2} - \frac{1}{2}\cos 2\theta$.)

$$
\begin{aligned}
\cos^2 2\theta &= \frac{1}{2} - \frac{1}{2}\cos 2\theta\\
2\cos^2 2\theta + \cos 2\theta - 1 &= 0\\
(2\cos 2\theta - 1)(\cos 2\theta + 1) &= 0\\
\cos 2\theta &= \frac{1}{2} \text{ or } -1\\
\cos 2\theta &= \frac{1}{2}: 2\theta = 60°, 300°, 420°, 660°,\\
&\text{so } \theta = 30°, 150°, 210°, 330°\\
\cos 2\theta &= -1: 2\theta = 180°, 540°, \text{so } \theta = 90°, 270°\\
\therefore\ \theta &= 30°, 90°, 150°, 210°, 270° \text{ or } 330°
\end{aligned}
$$

h $\cos 2\theta + 3 = 3\sin 2\theta$

(Hint: Write 3 as $3\cos^2\theta + 3\sin^2\theta$.)

$$
\begin{aligned}
\cos 2\theta + 3\cos^2\theta + 3\sin^2\theta &= 3\sin 2\theta\\
\cos^2\theta - \sin^2\theta + 3\cos^2\theta + 3\sin^2\theta &= 3\sin 2\theta\\
4\cos^2\theta + 2\sin^2\theta &= 6\sin\theta\cos\theta\\
4\cos^2\theta - 6\sin\theta\cos\theta + 2\sin^2\theta &= 0\\
2(2\cos\theta - \sin\theta)(\cos\theta - \sin\theta) &= 0\\
\tan\theta &= 2 \text{ or } \tan\theta = 1\\
\therefore\ \theta &= 45°, 63°26', 225° \text{ or } 243°
\end{aligned}
$$

8 Consider the equation $\tan\left(\frac{\pi}{4}+\theta\right)=3\tan\left(\frac{\pi}{4}-\theta\right)$.

a Show that $\tan^2\theta-4\tan\theta+1=0$.

b Hence use the quadratic formula to solve the equation for $0\leq\theta\leq\pi$.

$$\frac{1+\tan\theta}{1-\tan\theta}=\frac{3(1-\tan\theta)}{1+\tan\theta}$$
$$(1+\tan\theta)^2=3(1-\tan\theta)^2$$
$$1+2\tan\theta+\tan^2\theta=3-6\tan\theta+3\tan^2\theta$$
$$2\tan^2\theta-8\tan\theta+2=0$$
$$\tan^2\theta-4\tan\theta+1=0$$

$$\tan\theta=\frac{4\pm\sqrt{16-4}}{2}=\frac{4\pm2\sqrt{3}}{2}=2\pm\sqrt{3}$$
$$\tan\theta=2+\sqrt{3}:\theta=\frac{5\pi}{12}$$
$$\tan\theta=2-\sqrt{3}:\theta=\frac{\pi}{12}$$
$$\therefore\ \theta=\frac{\pi}{12}\text{ or }\frac{5\pi}{12}$$

9 Given the equation $2\cos x-1=2\cos 2x$:

a Show that $\cos x=\frac{1+\sqrt{5}}{4}$ or $\cos x=\frac{1-\sqrt{5}}{4}$.

b Hence solve the equation for $0\leq x\leq 2\pi$.

$$2\cos x-1=2(2\cos^2x-1)$$
$$2\cos x-1=4\cos^2x-2$$
$$4\cos^2x-2\cos x-1=0$$
$$\cos x=\frac{2\pm\sqrt{4+16}}{8}=\frac{2\pm\sqrt{20}}{8}=\frac{1\pm\sqrt{5}}{4}$$
$$\cos x=\frac{1+\sqrt{5}}{4}\ x=\frac{\pi}{5},\frac{9\pi}{5}$$
$$\cos x=\frac{1-\sqrt{5}}{4}:x=\frac{3\pi}{5},x=\frac{7\pi}{5}$$
$$\therefore\ x=\frac{\pi}{5},\frac{3\pi}{5},\frac{7\pi}{5}\text{ or }\frac{9\pi}{5}$$

10

a Show that $\sin(\alpha+\beta)\sin(\alpha-\beta)=\sin^2\alpha-\sin^2\beta$.

b Hence solve the equation $\sin^23\theta-\sin^2\theta=\sin 2\theta$, for $0\leq\theta\leq\pi$.

$$\begin{aligned}&\sin(\alpha+\beta)\sin(\alpha-\beta)\\&=(\sin\alpha\cos\beta+\cos\alpha\sin\beta)(\sin\alpha\cos\beta-\cos\alpha\sin\beta)\\&=\sin^2\alpha\cos^2\beta-\cos^2\alpha\sin^2\beta\\&=\sin^2\alpha(1-\sin^2\beta)-(1-\sin^2\alpha)\sin^2\beta\\&=\sin^2\alpha-\sin^2\alpha\sin^2\beta-\sin^2\beta+\sin^2\alpha\sin^2\beta\\&=\sin^2\alpha-\sin^2\beta\end{aligned}$$

$$\sin^23\theta-\sin^2\theta=\sin 2\theta$$
$$\sin(3\theta+\theta)\sin(3\theta-\theta)=\sin 2\theta,\text{ using part (a)}$$
$$\sin 4\theta\sin 2\theta=\sin 2\theta$$
$$\sin 2\theta(\sin 4\theta-1)=0$$
$$\sin 2\theta=0\text{ or }\sin 4\theta=1$$
$$\sin 2\theta=0:2\theta=0,\pi,2\pi,\text{so }\theta=0,\frac{\pi}{2},\pi$$
$$\sin 4\theta=1:4\theta=\frac{\pi}{2},\frac{5\pi}{2},\text{so }\theta=\frac{\pi}{8},\frac{5\pi}{8}$$
$$\therefore\ \theta=0,\frac{\pi}{8},\frac{\pi}{2},\frac{5\pi}{8}\text{ or }\pi$$

11

a Show that $\sin 3x = 3\sin x - 4\sin^3 x$.

b Hence solve the equation $\sin 3x + \sin 2x = \sin x$, for $0 \leq x \leq 2\pi$.

$$\begin{aligned}\sin 3x &= \sin(2x + x)\\ &= \sin 2x\cos x + \cos 2x \sin x\\ &= 2\sin x\cos^2 x + (1 - 2\sin^2 x)\sin x\\ &= 2\sin x(1 - \sin^2 x) + \sin x - 2\sin^3 x\\ &= 2\sin x - 2\sin^3 x + \sin x - 2\sin^3 x\\ &= 3\sin x - 4\sin^3 x\end{aligned}$$

$$\begin{aligned}3\sin x - 4\sin^3 x + 2\sin x\cos x &= \sin x\\ \sin x(3 - 4\sin^2 x + 2\cos x - 1) &= 0\\ \sin x(2 + 2\cos x - 4\sin^2 x) &= 0\\ \sin x\big(2 + 2\cos x - 4(1 - \cos^2 x)\big) &= 0\\ \sin x(4\cos^2 x + 2\cos x - 2) &= 0\\ \sin x(2\cos^2 x + \cos x - 1) &= 0\\ \sin x(2\cos x - 1)(\cos x + 1) &= 0\\ \sin x = 0 \text{ or } \cos x &= \frac{1}{2} \text{ or } \cos x = -1\\ \therefore x &= 0, \frac{\pi}{3}, \pi, \frac{5\pi}{3} \text{ or } 2\pi\end{aligned}$$

12 Use trigonometric identities and factoring to solve for $0° \leq \alpha \leq 360°$, giving solutions correct to the nearest minute where necessary.

a $\sec^2\alpha = 2\sec\alpha$

$$\begin{aligned}\frac{1}{\cos^2\alpha} &= \frac{2}{\cos\alpha}\\ 1 &= 2\cos\alpha\\ \cos\alpha &= \frac{1}{2}\\ \therefore \alpha &= 60° \text{ or } 300°\end{aligned}$$

b $\sec^2\alpha - \tan\alpha - 3 = 0$

$$\begin{aligned}\tan^2\alpha + 1 - \tan\alpha - 3 &= 0\\ \tan^2\alpha - \tan\alpha - 2 &= 0\\ (\tan\alpha - 2)(\tan\alpha + 1) &= 0\\ \tan\alpha &= 2 \text{ or } -1\\ \therefore \alpha &= 63°26', 135°, 243°26' \text{ or } 315\end{aligned}$$

c $\operatorname{cosec}^3 2\alpha = 4\operatorname{cosec} 2\alpha$

$$\begin{aligned}\operatorname{cosec}^3 2\alpha &= 4\operatorname{cosec} 2\alpha\\ \operatorname{cosec} 2\alpha(\operatorname{cosec}^2 2\alpha - 4) &= 0\\ \operatorname{cosec} 2\alpha &= \pm 2 \quad (\operatorname{cosec} 2\alpha \neq 0)\\ \sin 2\alpha &= \pm\frac{1}{2}\\ 2\alpha &= 30°, 150°, 210°, 330°, 390°, 510°, 570°, 690°\\ \alpha &= 15°, 75°, 105°, 165°, 195°, 255°, 285° \text{ or } 345°\end{aligned}$$

d $\sqrt{3}\operatorname{cosec}^2\left(\frac{\alpha}{2}\right) + \cot\left(\frac{\alpha}{2}\right) = \sqrt{3}$

$$\begin{aligned}\sqrt{3}\left(1 + \cot^2\left(\frac{\alpha}{2}\right)\right) + \cot\left(\frac{\alpha}{2}\right) &= \sqrt{3}\\ \sqrt{3}\cot^2\left(\frac{\alpha}{2}\right) + \cot\left(\frac{\alpha}{2}\right) &= 0\\ \cot\left(\frac{\alpha}{2}\right)\left(\sqrt{3}\cot\left(\frac{\alpha}{2}\right) + 1\right) &= 0\\ \cot\left(\frac{\alpha}{2}\right) = 0 \text{ or } \cot\left(\frac{\alpha}{2}\right) &= -\frac{1}{\sqrt{3}}\\ \cot\left(\frac{\alpha}{2}\right) = 0 &: \frac{\alpha}{2} = 90°, \text{so } \alpha = 180°\\ \cot\left(\frac{\alpha}{2}\right) = -\frac{1}{\sqrt{3}} &: \frac{\alpha}{2} = 120°, \text{so } \alpha = 240°\\ \therefore \alpha &= 180° \text{ or } 240°\end{aligned}$$

e $\sqrt{3}\operatorname{cosec}^2\alpha = 4\cot\alpha$

$$\begin{aligned}\frac{\sqrt{3}}{\sin^2\alpha} &= 4\frac{\cos\alpha}{\sin\alpha}\\ \sqrt{3} &= 4\cos\alpha\sin\alpha\\ \sqrt{3} &= 2\sin 2\alpha\\ \sin 2\alpha &= \frac{\sqrt{3}}{2}\\ 2\alpha &= 60°, 120°, 420°, 480°\\ \therefore \alpha &= 30°, 60°, 210° \text{ or } 240°\end{aligned}$$

f $\cot\alpha + 3\tan\alpha = 5\operatorname{cosec}\alpha$

$$\begin{aligned}\frac{\cos\alpha}{\sin\alpha} + \frac{3\sin\alpha}{\cos\alpha} &= \frac{5}{\sin\alpha}\\ \cos^2\alpha + 3\sin^2\alpha &= 5\cos\alpha\\ \cos^2\alpha + 3(1 - \cos^2\alpha) &= 5\cos\alpha\\ 3 - 2\cos^2\alpha &= 5\cos\alpha\\ 2\cos^2\alpha + 5\cos\alpha - 3 &= 0\\ (2\cos\alpha - 1)(\cos\alpha + 3) &= 0\\ \cos\alpha = \frac{1}{2} &\quad (\cos\alpha = -3 \text{ rejected})\\ \therefore \alpha &= 60° \text{ or } 300°\end{aligned}$$

Auxiliary Angle Method

Review

☐ Apply compound angle formula

a $\sin(x+\alpha)$

b $\cos(x+\alpha)$

c $\sin(x-\alpha)$

d $\cos(x-\alpha)$

☐ Simplify expressions using the Pythagorean Identity

a $\sin^2\theta+\cos^2\theta=\ldots\ldots$

b $R^2(\cos^2\theta+\sin^2\theta)=\ldots\ldots$

☐ Solve trigonometric equations involving tan

a $\tan\alpha=1$ for $\alpha\in[0,2\pi]$

b $\tan\alpha=\sqrt{3}$ for $\alpha\in[0,2\pi]$

c $\tan\alpha=\frac{1}{\sqrt{3}}$ for $\alpha\in[0,2\pi]$

☐ Identify quadrant

a $\sin\alpha>0,\cos\alpha>0$
Which quadrant is α in?

b $\sin\alpha>0,\cos\alpha<0$
Which quadrant is α in?

c $\sin\alpha<0,\cos\alpha>0$
Which quadrant is α in?

d $\sin\alpha<0,\cos\alpha<0$
Which quadrant is α in?

Investigation Superposition of waves.

Use Desmos to add two waves.

$$f(x) = a\sin x$$
$$g(x) = b\cos x$$
$$h(x) = f(x) + g(x)$$

Complete the sentence:

When you add two waves $y = a\sin x + b\cos x$, the resulting wave will have:

- Different a……
- Different p……
- Same p……

Therefore, we can write a sum of two waves as a single wave, with a new a…… and a p…… s……

$$y = a\sin x + b\cos x$$
$$= R\sin(x \pm \alpha) \text{ or } R\cos(x \pm \alpha)$$

α is called the **auxiliary** angle.

Auxiliary means “supporting”, in that it can help you solve trigonometric equations.

Form equations by comparing coefficients

Imagine that we are given $3\sin x + 4\cos x$ and asked to write it in the form $R\sin(x+\alpha)$.

If we expand $R\sin(x+\alpha)$, we get:

$$R\sin(x+\alpha) = R\sin x\cos\alpha + R\cos x\sin\alpha$$
$$3\sin x + 4\cos x = R\sin x\cos\alpha + R\cos x\sin\alpha$$

Therefore, by comparing coefficients:

$$3\sin x + 4\cos x = R\sin x\cos\alpha + R\cos x\sin\alpha$$
$$3 = R\cos\alpha$$
$$4 = R\sin\alpha$$

Forming these equations is the first step to solving for R and α.

By expanding and then comparing coefficients,
write two equations in terms of R and α for each:

a $5\sin x + 12\cos x = R\sin x\cos\alpha + R\cos x\sin\alpha$

b $\sin x - \cos x = R\sin(x+\alpha)$

c $3\sin x - 4\cos x = R\sin(x-\alpha)$

d $7\sin x - \sqrt{3}\cos x = R\sin(x-\alpha)$

e $-3\sin x + 4\cos x = R\cos(x+\alpha)$

f $\sqrt{3}\sin x + \cos x = R\cos(x-\alpha)$

ⓘ **Solve** simultaneous equations in the context of auxiliary angle method

Given $R\cos\alpha = x$, $R\sin\alpha = y$, we solve for R by squaring and using the Pythagorean identity.

(the question will usually tell you that $R > 0$)

$$R^2\cos^2\alpha = A^2$$
$$R^2\sin^2\alpha = y^2$$
$$R^2\cos^2\alpha + R^2\sin^2\alpha = x^2 + y^2$$
$$R^2(\cos^2\alpha + \sin^2\alpha) = x^2 + y^2$$
$$R^2 = x^2 + y^2$$
$$R = \sqrt{x^2 + y^2} \quad [R > 0]$$

The Pythagorean identity eliminates α, so an equation with R is left.

Solve for R.

a $R\cos\alpha = 3, \quad R\sin\alpha = 4$	**b** $R\cos\alpha = 1, \quad R\sin\alpha = 1$
c $R\cos\alpha = 5, \quad R\sin\alpha = 12$	**d** $R\cos\alpha = 1, \quad R\sin\alpha = \sqrt{3}$

Solve simultaneous equations in the context of auxiliary angle method

Given $R\cos\alpha = x$, $R\sin\alpha = y$, we solve for α by dividing sin by cos.

$$\frac{R\sin\alpha}{R\cos\alpha} = \frac{y}{x}$$

$$\tan\alpha = \frac{y}{x}$$

This eliminates R and leaves an equation with only one trig function: $\tan\alpha$.

To determine the quadrant, use the sign of $\cos\alpha$ and $\sin\alpha$.

Solve for α, assume $R > 0$.

a $R\cos\alpha = 1, \quad R\sin\alpha = \sqrt{3}$	**b** $R\cos\alpha = 1, \quad R\sin\alpha = 1$
c $R\cos\alpha = 1, \quad R\sin\alpha = -\sqrt{3}$	**d** $R\cos\alpha = -1, \quad R\sin\alpha = 1$

ⓘ Transforming to Auxiliary Angle Form

1. Expand $R\sin(x \pm \alpha)$ or $R\cos(x \pm \alpha)$ using the compound angle formula.
2. Equate coefficients of $\sin x$ and $\cos x$ to form equations for R and α.
3. Solve for R and α simultaneously.

   To find R: Use the Pythagorean identity.

   To find α: Divide the two equations to make a single equation in $\tan\alpha$.
   Identify the quadrant using the signs in step 2.

Example Transform to auxiliary angle form.

Express $y = \sin x - \sqrt{3}\cos x$ as $y = R\sin(x + \alpha)$ where $R > 0$ and $0 \leq \alpha < 2\pi$.

1. Expand:

$$R\sin(x + \alpha) = R\sin x\cos\alpha + R\cos x\sin\alpha$$

2. Equate coefficients:

$$\sin x - \sqrt{3}\cos x = R\sin x\cos\alpha + R\cos x\sin\alpha$$

$$R\cos\alpha = 1$$

$$R\sin\alpha = -\sqrt{3}$$

3. Find R:

$$R^2\cos^2\alpha + R^2\sin^2\alpha = 1^2 + \left(-\sqrt{3}\right)^2$$

$$R^2 = 1^2 + \left(-\sqrt{3}\right)^2$$

$$R = \sqrt{1^2 + \left(-\sqrt{3}\right)^2}$$

$$= 2$$

Find α:

$$\frac{R\sin\alpha}{R\cos\alpha} = -\frac{\sqrt{3}}{1}$$

$$\tan\alpha = -\sqrt{3}$$

$$A = \tan^{-1}\left(\sqrt{3}\right) = \frac{\pi}{3}$$

$$\sin\alpha < 0, \quad \cos\alpha > 0, \quad \therefore Q4$$

$$\alpha = 2\pi - \frac{\pi}{3} = \frac{5\pi}{3}$$

$$\therefore \sin x - \sqrt{3}\cos x = 2\sin\left(x + \frac{5\pi}{3}\right)$$

Practice

a Express $\sin x + \sqrt{3}\cos x$ in the form $R\sin(x+\alpha)$ where $R > 0$ and $0 \leq \alpha < 2\pi$.

1. Expand:

2. Equate coefficients:

3. Find R:

Find α:

$2\sin\left(x + \frac{\pi}{3}\right)$

b Express $\sqrt{2}\cos x + \sin x$ in the form $R\sin(x-\alpha)$ where $R > 0$ and $0 \leq \alpha < 2\pi$, giving α to 1 d.p.

$\sqrt{3}\sin(x - 5.3)$

c Express $3\sin x - 4\cos x$ in the form $R\cos(x-\alpha)$ where $R > 0$ and $0 \leq \alpha < 360°$, giving α to the nearest degree.

$5\cos(x - 143°)$

1 Find R and α exactly, if $R > 0$ and $0 \leq \alpha < 2\pi$, and:

a $R\sin\alpha = \sqrt{3}$ and $R\cos\alpha = 1$

$$R^2 = (\sqrt{3})^2 + 1^2 = 4$$
$$R = 2$$
$$\tan\alpha = \frac{\sqrt{3}}{1} = \sqrt{3}, \alpha \text{ in Q1}$$
$$\therefore R = 2, \alpha = \frac{\pi}{3}$$

b $R\sin\alpha = 3$ and $R\cos\alpha = 3$

$$R^2 = 3^2 + 3^2 = 18$$
$$R = 3\sqrt{2}$$
$$\tan\alpha = \frac{3}{3} = 1, \alpha \text{ in Q1}$$
$$\therefore R = 3\sqrt{2}, \alpha = \frac{\pi}{4}$$

2 Find R (exactly) and α (correct to the nearest minute), if $R > 0$ and $0° \leq \alpha < 360°$, and:

a $R\sin\alpha = 5$ and $R\cos\alpha = 12$

$$R^2 = 5^2 + 12^2 = 169$$
$$R = 13$$
$$\tan\alpha = \frac{5}{12}, \alpha \text{ in Q1}$$
$$\alpha = 22°37'$$
$$\therefore R = 13, \alpha \approx 22°37'$$

b $R\cos\alpha = 2$ and $R\sin\alpha = 4$

$$R^2 = 2^2 + 4^2 = 20$$
$$R = 2\sqrt{5}$$
$$\tan\alpha = \frac{4}{2} = 2, \alpha \text{ in Q1}$$
$$\alpha = 63°26'$$
$$\therefore R = 2\sqrt{5}, \alpha \approx 63°26'$$

3

a If $\cos x - \sin x = A\cos(x + \alpha)$, show that $A\cos\alpha = 1$ and $A\sin\alpha = 1$.

$$A\cos(x + \alpha) = A\cos x\cos\alpha - A\sin x\sin\alpha$$
$$A\cos\alpha = 1$$
$$A\sin\alpha = 1$$

b Find the positive value of A by squaring and adding.

$$A^2\cos^2\alpha + A^2\sin^2\alpha = 1^2 + 1^2 = 2$$
$$A^2 = 2$$
$$\therefore A = \sqrt{2}$$

c Find α, if $0 \leq \alpha < 2\pi$.

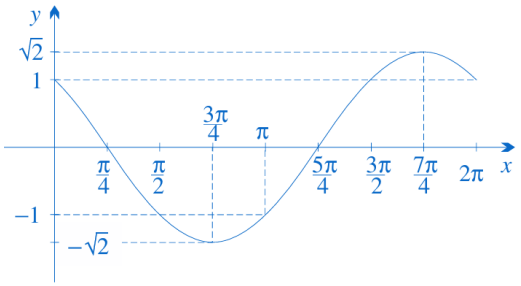
$$\tan\alpha = 1, \text{with } \sin\alpha > 0 \text{ and } \cos\alpha > 0 \text{ (Q1)}$$
$$\therefore \alpha = \frac{\pi}{4}$$

d Solve the equation $\cos x - \sin x = -1$, for $0 \leq x \leq 2\pi$.

$$\sqrt{2}\cos\left(x + \frac{\pi}{4}\right) = -1$$
$$\cos\left(x + \frac{\pi}{4}\right) = -\frac{1}{\sqrt{2}}$$
$$x + \frac{\pi}{4} = \frac{3\pi}{4}, \frac{5\pi}{4}$$
$$\therefore x = \frac{\pi}{2} \text{ or } \pi$$

e Write down the amplitude and period of $\cos x - \sin x$.
Hence sketch $y = \cos x - \sin x$, for $0 \leq x \leq 2\pi$.
Indicate on your sketch the line $y = -1$ and the solutions of the equation in part **d**.

Amplitude: $\sqrt{2}$ Period: 2π



4

a If $\sqrt{3}\cos x - \sin x = \mathrm{B}\cos(x + \theta)$, show that $\mathrm{B}\cos\theta = \sqrt{3}$ and $\mathrm{B}\sin\theta = 1$.

$$\mathrm{B}\cos(x + \theta) = \mathrm{B}\cos x\cos\theta - \mathrm{B}\sin x\sin\theta$$
$$\mathrm{B}\cos\theta = \sqrt{3}$$
$$\mathrm{B}\sin\theta = 1$$

b Find B, if $\mathrm{B} > 0$, by squaring and adding.

$$B^2\cos^2\theta + B^2\sin^2\theta = \left(\sqrt{3}\right)^2 + 1^2 = 4$$
$$\mathrm{B}^2 = 4$$
$$\therefore \mathrm{B} = 2$$

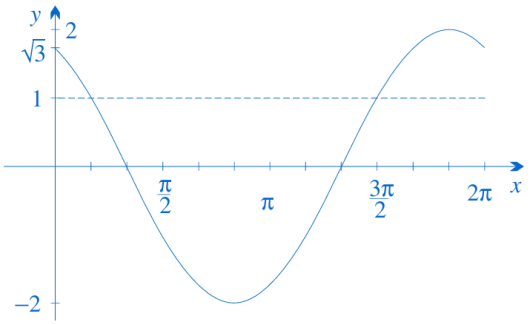
c Find θ, if $0 \leq \theta < 2\pi$.

$$\tan\theta = \frac{1}{\sqrt{3}}, \theta \text{ in Q1}$$
$$\therefore \theta = \frac{\pi}{6}$$

d Solve the equation $\sqrt{3}\cos x - \sin x = 1$, for $0 \leq x \leq 2\pi$.

$$2\cos\left(x + \frac{\pi}{6}\right) = 1$$
$$\cos\left(x + \frac{\pi}{6}\right) = \frac{1}{2}$$
$$x + \frac{\pi}{6} = \frac{\pi}{3}, \frac{5\pi}{3}$$
$$\therefore x = \frac{\pi}{6} \text{ or } \frac{3\pi}{2}$$

e Sketch $\mathrm{y} = \sqrt{3}\cos x - \sin x$, for $0 \leq x \leq 2\pi$.
On the same diagram, sketch the line $\mathrm{y} = 1$.
Indicate on your diagram the solutions of the equation in part **d**.



5 Let $4\sin x - 3\cos x = \mathrm{A}\sin(x - \alpha)$, where $\mathrm{A} > 0$ and $0° \leq \alpha < 360°$.

a Show that $\mathrm{A}\cos\alpha = 4$ and $\mathrm{A}\sin\alpha = 3$.

$$\mathrm{A}\sin(x - \alpha) = \mathrm{A}\sin x\cos\alpha - \mathrm{A}\cos x\sin\alpha$$
$$\mathrm{A}\cos\alpha = 4$$
$$\mathrm{A}\sin\alpha = 3$$

b Show that $\mathrm{A} = 5$ and $\alpha = \tan^{-1}\left(\frac{3}{4}\right)$.

$$\mathrm{A}^2 = 4^2 + 3^2 = 25,$$
$$\mathrm{A} = 5$$
$$\tan\alpha = \frac{3}{4}, \text{so } \alpha = \tan^{-1}\left(\frac{3}{4}\right)$$

c Hence solve the equation $4\sin x - 3\cos x = 5$, for $0° \leq x \leq 360°$.
Give the solution(s) correct to the nearest minute.

$$5\sin(x - \alpha) = 5$$
$$\sin(x - \alpha) = 1$$
$$x - \alpha = 90°$$
$$x = 90° + \tan^{-1}\left(\frac{3}{4}\right) = 90° + 36°52' = 126°52'$$
$$\therefore x \approx 126°52'$$

6 Consider the equation $2\cos x + \sin x = 1$.

a Let $2\cos x + \sin x = \mathrm{B}\cos(x - \theta)$, where $\mathrm{B} > 0$ and $0° \leq \theta < 360°$.

Show that $\mathrm{B} = \sqrt{5}$ and $\theta = \tan^{-1}\left(\frac{1}{2}\right)$.

$$\begin{aligned}
\mathrm{B}\cos(x - \theta) &= \mathrm{B}\cos x\cos\theta + \mathrm{B}\sin x\sin\theta \\
\mathrm{B}\cos\theta &= 2, \mathrm{B}\sin\theta = 1 \\
\mathrm{B}^2 &= 4 + 1 = 5, \\
\mathrm{B} &= \sqrt{5} \\
\tan\theta &= \frac{1}{2}, \\
\theta &= \tan^{-1}\left(\frac{1}{2}\right)
\end{aligned}$$

b Hence find, correct to the nearest minute where necessary, the solutions of the equation, for $0° \leq x \leq 360°$.

$$\begin{aligned}
\sqrt{5}\cos(x - \theta) &= 1 \\
\cos(x - \theta) &= \frac{1}{\sqrt{5}} \\
\theta &= \tan^{-1}\left(\frac{1}{2}\right) = 26°34' \\
x - 26°34' &= 63°26', 296°34' \\
x &= 90° \text{ or } 323°8' \\
\therefore x &= 90° \text{ or } x \approx 323°8'
\end{aligned}$$

7 Let $\cos x - 3\sin x = \mathrm{D}\cos(x + \varphi)$, where $\mathrm{D} > 0$ and $0° \leq \varphi < 360°$.

a Show that $\mathrm{D} = \sqrt{10}$ and $\varphi = \tan^{-1}3$.

$$\begin{aligned}
\mathrm{D}\cos(x + \varphi) &= \mathrm{D}\cos x\cos\varphi - \mathrm{D}\sin x\sin\varphi \\
\mathrm{D}\cos\varphi &= 1, \mathrm{D}\sin\varphi = 3 \\
\mathrm{D}^2 &= 1 + 9 = 10, \\
\mathrm{D} &= \sqrt{10} \\
\tan\varphi &= 3, \\
\varphi &= \tan^{-1}3 \text{ (Q1)}
\end{aligned}$$

b Hence solve $\cos x - 3\sin x = 3$, for $0° \leq x \leq 360°$.

Give the solutions correct to the nearest minute where necessary.

$$\begin{aligned}
\sqrt{10}\cos(x + \varphi) &= 3 \\
\cos(x + \varphi) &= \frac{3}{\sqrt{10}} \\
\varphi &= \tan^{-1}3 = 71°34' \\
x + 71°34' &= 18°26', 341°34', 360° + 18°26' \\
x &= -53°8' \text{(rejected)}, 270°, 306°52' \\
\therefore x &= 270° \text{ or } x \approx 306°52'
\end{aligned}$$

8 Consider the equation $\sqrt{5}\sin x + 2\cos x = -2$.

a Transform the LHS into the form $C\sin(x + \alpha)$, where $C > 0$ and $0° \leq \alpha < 360°$.

$$C\sin(x + \alpha) = C\sin x\cos\alpha + C\cos x\sin\alpha$$
$$C\cos\alpha = \sqrt{5}, C\sin\alpha = 2$$
$$C^2 = 5 + 4 = 9,$$
$$\text{so } C = 3$$
$$\tan\alpha = \frac{2}{\sqrt{5}},$$
$$\alpha = \tan^{-1}\left(\frac{2}{\sqrt{5}}\right) = 41°49' \text{ (Q1)}$$
$$\therefore \sqrt{5}\sin x + 2\cos x = 3\sin\left(x + \tan^{-1}\left(\frac{2}{\sqrt{5}}\right)\right)$$

b Find, correct to the nearest minute where necessary, the solutions of the equation, for $0° \leq x \leq 360°$.

$$3\sin(x + 41°49') = -2$$
$$\sin(x + 41°49') = -\frac{2}{3}$$
$$x + 41°49' = 221°49', \text{or } 318°11'$$
$$x = 180° \text{ or } 276°23'$$

9 Solve each equation, for $0° \leq x \leq 360°$, by transforming the LHS into a single-term sine or cosine function. Give solutions correct to the nearest minute.

a $3\sin x + 5\cos x = 4$

$$3\sin x + 5\cos x = \sqrt{34}\sin(x + \alpha), \text{where } \tan\alpha = \frac{5}{3}, \alpha = 59°2'$$
$$\sqrt{34}\sin(x + 59°2') = 4$$
$$\sin(x + 59°2') = \frac{4}{\sqrt{34}}$$
$$x + 59°2' = 43°19' \text{ (rejected)}, 136°41', 403°19'$$
$$x = 77°39' \text{ or } 344°17'$$

b $6\sin x - 5\cos x = 7$

$$6\sin x - 5\cos x = \sqrt{61}\sin(x - \alpha), \text{where } \tan\alpha = \frac{5}{6}, \alpha = 39°48'$$
$$\sqrt{61}\sin(x - 39°48') = 7$$
$$\sin(x - 39°48') = \frac{7}{\sqrt{61}} \approx 0.896$$
$$x - 39°48' = 63°37', 116°23'$$
$$x = 103°29' \text{ or } 156°8'$$

10 Find A and α exactly, if $A > 0$ and $0 \leq \alpha < 2\pi$, and:

a $A\sin\alpha = 1$ and $A\cos\alpha = -\sqrt{3}$

$$A^2 = 1^2 + \left(-\sqrt{3}\right)^2 = 4,$$
$$A = 2$$
$$\tan\alpha = \frac{1}{-\sqrt{3}} = -\frac{1}{\sqrt{3}}$$
$$\sin\alpha > 0, \cos\alpha < 0, \text{so } \alpha \text{ in Q2}$$
$$\therefore A = 2, \alpha = \frac{5\pi}{6}$$

b $A\cos\alpha = -5$ and $A\sin\alpha = -5$

$$A^2 = (-5)^2 + (-5)^2 = 50, A = 5\sqrt{2}$$
$$\tan\alpha = \frac{-55}{-5} = 1$$
$$\sin\alpha < 0, \cos\alpha < 0, \text{so } \alpha \text{ in Q3}$$
$$\therefore A = 5\sqrt{2}, \alpha = \frac{5\pi}{4}$$

11 Find A (exactly) and α (correct to the nearest minute), if $A > 0$ and $0° \leq \alpha < 360°$, and:

a $A\cos\alpha = 5$ and $A\sin\alpha = -4$

$$A^2 = 5^2 + (-4)^2 = 41, A = \sqrt{41}$$
$$\tan\alpha = -\frac{4}{5}$$
$$\sin\alpha < 0, \cos\alpha > 0, \text{so } \alpha \text{ in Q4}$$
$$\alpha = 360° - 38°40' = 321°20'$$
$$\therefore A = \sqrt{41}, \alpha \approx 321°20'$$

b $A\sin\alpha = -11$ and $A\cos\alpha = -2$

$$A^2 = (-2)^2 + (-11)^2 = 125, A = 5\sqrt{5}$$
$$\tan\alpha = \frac{-11}{-2} = \frac{11}{2}$$
$$\sin\alpha < 0, \cos\alpha < 0, \text{so } \alpha \text{ in Q3}$$
$$\alpha = 180° + 79°42' = 259°42'$$
$$\therefore A = 5\sqrt{5}, \alpha \approx 259°42'$$

12

a Express $\sqrt{3}\cos x + \sin x$ in the form $A\cos(x + \theta)$, where $A > 0$ and $0 < \theta < 2\pi$.

$$\sqrt{3}\cos x + \sin x = A\cos x\cos\theta - A\sin x\sin\theta$$
$$A\cos\theta = \sqrt{3}, A\sin\theta = -1$$
$$A = 2, \tan\theta = -\frac{1}{\sqrt{3}}, \theta \text{ in Q4}$$
$$\theta = 2\pi - \frac{\pi}{6} = \frac{11\pi}{6}$$
$$\therefore \sqrt{3}\cos x + \sin x = 2\cos\left(x + \frac{11\pi}{6}\right)$$

b Hence solve $\sqrt{3}\cos x + \sin x = 1$, for $0 \leq x < 2\pi$.

$$2\cos\left(x + \frac{11\pi}{6}\right) = 1$$
$$\cos\left(x + \frac{11\pi}{6}\right) = \frac{1}{2}$$
$$x + \frac{11\pi}{6} = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3}$$
$$x = -\frac{3\pi}{2}(\text{rejected}), -\frac{\pi}{6}(\text{rejected}), \frac{\pi}{2}, \frac{11\pi}{6}$$
$$\therefore x = \frac{\pi}{2} \text{ or } \frac{11\pi}{6}$$

13

a Express $\cos x - \sin x$ in the form $B\sin(x + \alpha)$, where $B > 0$ and $0 < \alpha < 2\pi$.

$$\cos x - \sin x = B\sin x\cos\alpha + B\cos x\sin\alpha$$
$$B\sin\alpha = 1, B\cos\alpha = -1$$
$$B = \sqrt{2},$$
$$\tan\alpha = -1,$$
$$\alpha = \frac{3\pi}{4} \text{ (Q2)}$$
$$\therefore \cos x - \sin x = \sqrt{2}\sin\left(x + \frac{3\pi}{4}\right)$$

b Hence solve $\cos x - \sin x = 1$, for $0 \leq x < 2\pi$.

$$\sqrt{2}\sin\left(x + \frac{3\pi}{4}\right) = 1$$
$$\sin\left(x + \frac{3\pi}{4}\right) = \frac{1}{\sqrt{2}}$$
$$x + \frac{3\pi}{4} = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{\pi}{4} + 2\pi$$
$$x = -\frac{\pi}{2}(\text{rejected}), 0, \frac{3\pi}{2}$$
$$\therefore x = 0 \text{ or } \frac{3\pi}{2}$$

14

a Express $\sin x - \sqrt{3}\cos x$ in the form $C\sin(x + \beta)$, where $C > 0$ and $0 < \beta < 2\pi$.

$$\sin x - \sqrt{3}\cos x = C\sin x\cos\beta + C\cos x\sin\beta$$
$$C\cos\beta = 1, C\sin\beta = -\sqrt{3}$$
$$C = 2,$$
$$\tan\beta = -\sqrt{3},$$
$$\beta = \frac{5\pi}{3} \text{ (Q4)}$$
$$\therefore \sin x - \sqrt{3}\cos x = 2\sin\left(x + \frac{5\pi}{3}\right)$$

b Hence solve $\sin x - \sqrt{3}\cos x = -1$, for $0 \leq x < 2\pi$.

$$2\sin\left(x + \frac{5\pi}{3}\right) = -1$$
$$\sin\left(x + \frac{5\pi}{3}\right) = -\frac{1}{2}$$
$$x + \frac{5\pi}{3} = \frac{7\pi}{6}, \frac{11\pi}{6}, \frac{19\pi}{2}$$
$$x = -\frac{\pi}{2} \text{ (rejected)}, \frac{\pi}{6}, \frac{3\pi}{2}$$
$$\therefore x = \frac{\pi}{6} \text{ or } \frac{3\pi}{2}$$

15

a Express $-\cos x - \sin x$ in the form $D\cos(x - \varphi)$, where $D > 0$ and $0 < \varphi < 2\pi$.

$$-\cos x - \sin x = D\cos x\cos\varphi + D\sin x\sin\varphi$$
$$D\cos\varphi = -1, D\sin\varphi = -1$$
$$D = \sqrt{2},$$
$$\tan\varphi = 1,$$
$$\varphi = \frac{5\pi}{4} \text{ (Q3)}$$
$$\therefore -\cos x - \sin x = \sqrt{2}\cos\left(x - \frac{5\pi}{4}\right)$$

b Hence solve $-\cos x - \sin x = 1$, for $0 \leq x < 2\pi$.

$$\sqrt{2}\cos\left(x - \frac{5\pi}{4}\right) = 1$$
$$\cos\left(x - \frac{5\pi}{4}\right) = \frac{1}{\sqrt{2}}$$
$$x - \frac{5\pi}{4} = \frac{\pi}{4}, \frac{7\pi}{4}, -\frac{\pi}{4}$$
$$x = \frac{3\pi}{2}, 3\pi \text{ (rejected)}, \pi$$
$$\therefore x = \pi \text{ or } \frac{3\pi}{2}$$

16

a Express $-3\sin x - 4\cos x$ in the form $S\cos(x - \beta)$, where $S > 0$ and $0 < \beta < 2\pi$. (Write β correct to four decimal places.)

$$-3\sin x - 4\cos x = S\cos x\cos\beta + S\sin x\sin\beta$$
$$S\cos\beta = -4, S\sin\beta = -3$$
$$S = 5, \tan\beta = \frac{3}{4}, \beta \text{ in Q3}$$
$$\beta = \pi + 0.6435 = 3.7851$$
$$\therefore -3\sin x - 4\cos x = 5\cos(x - 3.7851)$$

b Hence solve $-3\sin x - 4\cos x = 2$, for $0 \leq x < 2\pi$. Give the solutions correct to two decimal places.

$$5\cos(x - 3.7851) = 2$$
$$\cos(x - 3.7851) = 0.4$$
$$x - 3.7851 = 1.1593, -1.1593$$
$$x \approx 4.94 \text{ or } 2.63$$

17

a Given the equation $\sin\theta + \cos\theta = \cos 2\theta$, show that $\tan\theta = -1$ or $\cos\theta - \sin\theta = 1$.

$$\begin{aligned}\sin\theta + \cos\theta &= \cos^2\theta - \sin^2\theta\\ \sin\theta + \cos\theta &= (\cos\theta - \sin\theta)(\cos\theta + \sin\theta)\\ (\sin\theta + \cos\theta)(1 - \cos\theta + \sin\theta) &= 0\\ \sin\theta + \cos\theta &= 0 \text{ or } \cos\theta - \sin\theta = 1\\ \tan\theta &= -1 \text{ or } \cos\theta - \sin\theta = 1\end{aligned}$$

b Hence solve $\sin\theta + \cos\theta = \cos 2\theta$, for $0 \le \theta < 2\pi$.

$$\begin{aligned}\tan\theta = -1&: \theta = \frac{3\pi}{4}, \frac{7\pi}{4}\\ \cos\theta - \sin\theta = 1&: \sqrt{2}\cos\left(\theta + \frac{\pi}{4}\right) = 1\\ \cos\left(\theta + \frac{\pi}{4}\right) &= \frac{1}{\sqrt{2}}\\ \theta + \frac{\pi}{4} &= \frac{\pi}{4}, \frac{7\pi}{4}\\ \theta &= 0, \frac{3\pi}{2}\\ \therefore \theta &= 0, \frac{3\pi}{4}, \frac{3\pi}{2} \text{ or } \frac{7\pi}{4}\end{aligned}$$

18 Solve, for $0 \le x \le 2\pi$:

a $\sqrt{3}\sin 2x - \cos 2x = 2$

$$\begin{aligned}\sqrt{3}\sin 2x - \cos 2x &= 2\\ 2\sin\left(2x - \frac{\pi}{6}\right) &= 2\\ \sin\left(2x - \frac{\pi}{6}\right) &= 1\\ 2x - \frac{\pi}{6} &= \frac{\pi}{2}\\ 2x &= \frac{2\pi}{3}\\ \therefore x &= \frac{\pi}{3} \text{ or } \frac{4\pi}{3}\end{aligned}$$

b $\sin 4x + \cos 4x = 1$

$$\begin{aligned}\sin 4x + \cos 4x &= 1\\ \sqrt{2}\sin\left(4x + \frac{\pi}{4}\right) &= 1\\ \sin\left(4x + \frac{\pi}{4}\right) &= \frac{1}{\sqrt{2}}\\ 4x + \frac{\pi}{4} &= \frac{\pi}{4}, \frac{3\pi}{4}, \frac{9\pi}{4}, \frac{11\pi}{4}, \frac{17\pi}{4}, \frac{19\pi}{4}, \frac{25\pi}{4}, \frac{27\pi}{4}\\ 4x &= 0, \frac{\pi}{2}, 2\pi, \frac{5\pi}{2}, 4\pi, \frac{9\pi}{2}, 6\pi, \frac{13\pi}{2}\\ \therefore x &= 0, \frac{\pi}{8}, \frac{\pi}{2}, \frac{5\pi}{8}, \pi, \frac{9\pi}{8}, \frac{3\pi}{2}, \frac{13\pi}{8} \text{ or } 2\pi\end{aligned}$$

19

a Show that $\sin x - \cos x = \sqrt{2}\sin\left(x - \frac{\pi}{4}\right)$.

b Hence sketch the graph of $y = \sin x - \cos x$, for $0 \leq x \leq 2\pi$.

c Use your sketch to determine the values of x in the domain $0 \leq x \leq 2\pi$ for which $\sin x - \cos x > 1$.

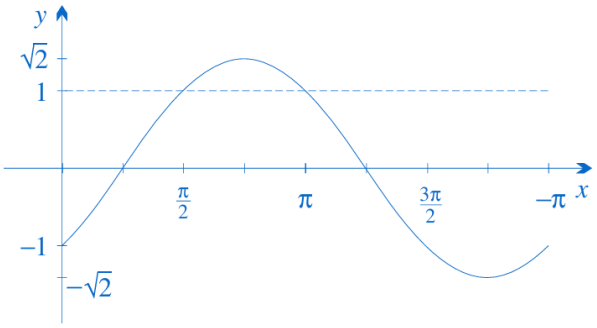
$\sin x - \cos x = R\sin(x - \alpha)$

$= R\sin x\cos\alpha - R\cos x\sin\alpha$

$R\cos\alpha = 1, R\sin\alpha = 1$

$R = \sqrt{2}, \alpha = \tan^{-1}1 = \frac{\pi}{4}$ (Q1)

$\therefore \sin x - \cos x = \sqrt{2}\sin\left(x - \frac{\pi}{4}\right)$



$\sqrt{2}\sin\left(x - \frac{\pi}{4}\right) > 1$

$\sin\left(x - \frac{\pi}{4}\right) > \frac{1}{\sqrt{2}}$

using graph: $\frac{\pi}{2} < x < \pi$

20 The depth of water in a harbour is modelled by

$$d = 5 + 3\sin t + 4\cos t$$

where d is in metres and t is time in hours after midnight.

Express $3\sin t + 4\cos t$ in the form $R\sin(t + \alpha)$, hence find the maximum depth of water (maximum value of d) and the time after midnight when this maximum first occurs.

$3\sin t + 4\cos t = R\sin(t + \alpha) = R\cos\alpha\sin t + R\sin\alpha\cos t$

Comparing coefficients:

$R\cos\alpha = 3 \ldots (1)$

$R\sin\alpha = 4 \ldots (2)$

$(1)^2 + (2)^2$: $R^2 = 9 + 16 = 25$,

$so\ R = 5$

$(2) \div (1)$: $\tan\alpha = \frac{4}{3}$,

$\alpha = \tan^{-1}\left(\frac{4}{3}\right)$

$3\sin t + 4\cos t = 5\sin\left(t + \tan^{-1}\left(\frac{4}{3}\right)\right)$

$d = 5 + 5\sin\left(t + \tan^{-1}\left(\frac{4}{3}\right)\right)$

Maximum depth occurs when $\sin\left(t + \tan^{-1}\left(\frac{4}{3}\right)\right) = 1$

$\therefore$ Maximum depth $= 5 + 5 = 10$ metres

For maximum: $t + tan^{-1}\left(\frac{4}{3}\right) = \frac{\pi}{2}$

$t = \frac{\pi}{2} - tan^{-1}\left(\frac{4}{3}\right)$

$t \approx 0.6435$ hours

$t \approx 38.6$ minutes

$\therefore$ Maximum first occurs at approximately 12:39 am (or 0.644 hours after midnight)